SUPPLEMENT: PROOFS, SERVING STACK, PRE-REGISTRATION, POWER, PUBLIC TRAJECTORIES, LOGS

Ghaisan Khoirul Badruzaman
Independent Researcher

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S1. Proofs of Propositions 1 to 3

The propositions, their assumptions and their notation are stated in Section 3 of the paper. This section gives the full proofs.

Proposition 1. A candidate is accepted with probability $1-a$. With probability $1-a^n$ at least one is accepted, and by exchangeability each accepted candidate is correct with probability q, whatever the number accepted. With probability a^n none is, and a rejected candidate is correct with probability $p(1-\alpha)/a$. So $P_n = (1-a^n)q + a^{n-1}p(1-\alpha) = q - a^{n-1}\big(aq - p(1-\alpha)\big)$. At $n=1$ this equals p, so $aq - p(1-\alpha) = q - p$ and $P_n = q - (q-p)a^{n-1}$. Finally $q/(1-q) = p\alpha/\big((1-p)\beta\big)$.

Proposition 2. Since $0 \le a < 1$, $P_n - p = (q-p)(1-a^{n-1})$ moves monotonically from 0 towards $q-p$, and for $\alpha \ge \beta > 0$, $q - p = p(1-p)(\alpha-\beta)/\big(p\alpha + (1-p)\beta\big) \le p(1-p)(\alpha-\beta)/\big((1-p)\beta\big) = p(\mathrm{LR}-1)$.

Proposition 3. *Statement.* Let a fraction b of an author's errors be blind spots that a judge sharing its information always misses, let any judge miss the other errors with probability m_0, and let an independent judge miss blind-spot errors with probability $m_1 < 1$. Then independent judgement lets through a fraction $\rho = \big(b\,m_1 + (1-b)\,m_0\big)/\big(b + (1-b)\,m_0\big) < 1$ whenever $b > 0$ of the errors self-judgement lets through. *Proof.* An error escapes when the author errs, with probability ε, and the judge misses it. Blind-spot errors escape a self-judge always and an independent judge with probability m_1; other errors escape either with probability m_0. The escaped-error rates are $\varepsilon[b + (1-b)m_0]$ and $\varepsilon[b\,m_1 + (1-b)m_0]$, whose ratio ρ is below one exactly when $b\,m_1 < b$, that is when $b > 0$ and $m_1 < 1$.

S2. Serving stack and the outer level

The inner level consists of the competition's harness and wheelhouse (Kaggle dataset `metric/gemma-4-developer-agent-wheelhouse`; the host notebook's image `gcr.io/kaggle-private-byod/python@sha256:37c64f7d...`) and the model `google/gemma-4/Other/gemma-4-31b-it-qat-w4a16-ct/2` (a 4-bit QAT checkpoint) served by vLLM on four L4 GPUs with `max_model_len` 32,768 (harness README). The median decode speed was 34.2 to 34.7 tokens per second in three 8-minute runs on the four L4s (r0014, r0017, r0018; read from their traces, which are not released), so a 4,096-token thinking turn costs about two minutes. The outer level is Claude Code 2.1.x, which ran Claude Opus 5.5 until 2026-10-10 and Claude Fable 5.1 from the 2026-10-10 relaunch. The five single-loop configurations of the patch pool and the baseline r0010 ran under harness version swegemma 0.2.7. Every run from r0044 on used 0.2.11 with a rollout watchdog added after the model-server hang of the first pipeline run. The `ledger.py compare` command refuses to pair runs across harness versions (the ledger's one-instrument rule), which is why the development result for time-budget governance (r0049 against r0010) is reported as descriptive. The P6 floor arm (r0054) ran on a single NVIDIA A100-SXM4-40GB (tensor-parallel size 1) with the same wheelhouse, vLLM 0.19.1 and Python 3.12.16. It reached a decoding throughput of about 43 tokens per second against 34 on the four L4s. Its

gate row against r0045 records every instrument difference (gpu_count, gpu_model, python, rollout_sha) and pairs the two runs by task id (`ledger.py compare -allow-diff`, introduced on 2026-11-04 alongside a failing test and a verifier review). Four of the floor arm's eight exclusive wins ran 218 to 247 s with 25 to 49 calls under the 4-minute cap. The slower decoder would likely have truncated them, so the faster decoder favours the floor arm. Because of this advantage and the unmeasured test-execution speed of the two sandboxes, the registered test on this pair ($p = 0.32$, gate row of 2026-11-04) is read as descriptive, and its direction is conservative for the system arm. In the parser-safe edit form the agent runs the single command `python3 /tmp/axi_apply.py «'AXI_EDIT'`. Its body holds the file path on the first line and then a block `«««« SEARCH` (the exact current lines) `=======` (their replacement) `»»»» REPLACE`. The applier refuses when the search block is absent or not unique and writes nothing. This edit form replaces the structured call `edit_file(path, old_string, new_string)`, which the harness's tool-call parser rejects whenever a field is missing from the model's JSON (601 of 2,057 calls in our traces, analysis parser-tax).

S3. Pre-registration timeline and the patch pool

The commits of the P5 pre-registration, its amendments and the interim look have these UTC times (git committer times of the campaign repository at the frozen commit): 438bd84 2026-10-04 10:22:00 UTC; 1e25bbf 2026-10-06 01:12:35 UTC; ed325c1 2026-10-06 01:21:00 UTC; 7776fed 2026-10-06 01:26:33 UTC; b69828f 2026-10-06 01:29:36 UTC; e2bd275 2026-10-06 01:45:27 UTC; 6c20b64 2026-10-06 01:56:33 UTC; 7c71f6f 2026-10-06 02:13:18 UTC; b1194c9 2026-10-06 03:03:00 UTC (interim report); d12c57d 2026-10-06 03:03:49 UTC (A8); f55543f 2026-10-06 08:03:54 UTC. A1 to A7 precede the interim's report commit (b1194c9). A8 is an operational amendment that came 49 s after it and before any fresh-judge verdict existed (a 600 s limit and four concurrent requests for the fresh judge, with verdicts independent of order and batching). A9 discloses the interim (its ledger row e0001 is timestamped 06:17:20 UTC, commit a5b0eec) and sets the strata: H6 is confirmatory on every stratum, and E3-a and H3 are confirmatory on patches judged after A9 and exploratory on pool v1. SHA-256 of P5, P6 and P7 are published in the Kaggle dataset `ghaisank/axi-preregistration-hashes` (SHA256SUMS lists the hashing clock, one to two seconds after each commit; the line for P6's amendment A4 was added on 2026-11-09, after the floor arm's outcome, although the amendment was committed on 2026-11-04T13:32:47Z, before that arm's first ledger row at 19:00Z). The texts of P5, P6 and P7 with their amendments are in `ghaisank/axi-paper-release` (preregistration/). The commits stay in the private campaign repository; the public evidence of timing is the version history of the hash dataset.

The patch pool (pool v1) holds the 182 non-empty patches that five single-loop configurations produced on 60 of the 71 development tasks (r0005, r0006, r0007, r0009, r0010; 89 correct, 93 wrong). P5 defined the pool as every non-empty development patch up to the freeze (2026-11-05). The fresh judge ran once (runs/judge-01, 2026-10-10), and a second judge run did not fit the GPU quota, so no later patch was judged by any of the four checks and the analysis stops at pool v1. This is a deviation from P5: the confirmatory stratum that A9 defines for E3-a and H3 (patches judged after A9) was never formed, so E3-a and H3 are exploratory and only H6 is confirmatory. The interim look (e0001, before the fresh-judge rule was final) gave self-claim LR 1.74 [1.37, 2.30], own reproduction 1.13 [1.01, 1.27] and nearby tests 1.08 [1.02, 1.15]. These are the same values as the final row e0002 for those three checks because their verdicts were unchanged by A8 and A9. The session-exit analysis finds that 55 of the 182 patches come from sessions that timed out (e3-terminal-01, 2026-11-05, run post hoc after a review objection). All 55 are wrong, and 54 of them score "no claim"; the one claim is a `submit_patch` call in a session that then ran to the cap (dev-02/floor-v0, fastapi_14791). The registered pooled ratios therefore mix judgement with finishing in time. Table 1's last block repeats the analysis on the 127 patches from sessions that finished (intervals from 10,000 bootstrap draws over patches, seed 20261105). Each check's α, β and LR are computed on the patches where its verdict is defined (own reproduction 107, nearby tests 132, the other two 182). The E3-a ratios therefore compare non-nested subsets, which is why Section 4 of the paper also recomputes the self-claim on those subsets (analysis e3-combo-01, exploratory: 10,000 bootstrap draws over tasks, seed 20261004). A conjunction accepts when every member accepts and is defined where every member is.

S4. Power of the paired tests

`ledger.py mde` is a simulation-based planning aid (one replicate, one-sided $\alpha = 0.10$, 80% power). At the planning base rate of 0.20 it gives a minimum detectable lift of 0.16 at 62 tasks and 0.24 at 32 tasks, and at the base rates observed in the control arms it gives 0.18 for the 71 development tasks (base 0.44, r0049) and 0.26 for the 62 AXI-Fresh tasks (base 0.74, r0045). The simulation draws each task's solve probability from a Beta distribution with the stated mean, adds the lift to every task (capped at 1) and samples both arms independently. At a base rate of 0.74 the

cap makes the detectable lift approach $1 - 0.74$, so the AXI-Fresh figure is a ceiling of the design. A reader should use the observed paired discordance instead (19 of 62 on that set). A lift below these figures is likely to go undetected.

S5. Self-claim and own reproduction on other agents' public trajectories (exploratory)

The rules are those frozen in P5 and mapped to each trajectory format (analysis e3-public). The self-claim is the agent's final submission declaring the fix done. Own reproduction is a script the agent wrote that passed after its last edit. Its result is read from the exit code where the format records one (OpenHands) and from the run's output where it does not (SWE-agent, as the frozen rule allows). Documentation-only writes do not count as edits. Sample frames are seeded subsets of row groups with 150 resolved and 150 unresolved instances per set. Outcomes are the datasets' own resolved flags. Intervals are 90% bootstrap intervals over tasks with 10,000 draws. "Judged" counts the sampled instances on which the check's verdict is defined. Own reproduction is undefined where the agent wrote no test (OpenHands 23 of 300; SWE-agent 133 of 300).

check	judged	sensitivity α	false acceptance β	LR
self-claim, OpenHands (Qwen3-Coder-480B)	300	0.95	0.79	1.20 (1.12–1.30)
own reproduction, OpenHands	277	0.98	0.91	1.08 (1.03–1.13)
self-claim, SWE-agent (Llama 3.1)	300	0.92	0.57	1.62 (1.44–1.85)
own reproduction, SWE-agent	167	0.84	0.54	1.56 (1.31–1.92)

Table S1: in-task checks on public trajectories of other agents; LR with its 90% interval.

S6. How the study was run (logs)

The experiments and the paper's text were produced by a Claude Code agent under fixed rules over five weeks (2026-10-04 to 2026-11-07, the span of the counted events): a gate (the 71 development tasks plus the paired sign test) no helper agent could edit, a 34-task holdout no agent could read, a reviewer that never reviewed its own work, and the ledger from which every run, leaderboard and test number is recomputed (`ledger.py verify`). The script `count_events.py` counts, up to 2026-11-08, 356 inputs from the author, 30 context compactions and 694 actions blocked by hooks (570) or script refusals (124) in the session transcripts (`events-2026-11-08.json`). After each compaction the agent recovered its goal from a frozen copy. The event counts are released in the dataset `ghaisank/axi-paper-release`; the logs they are counted from stay private because they hold personal data.

S7. The five-role pipeline

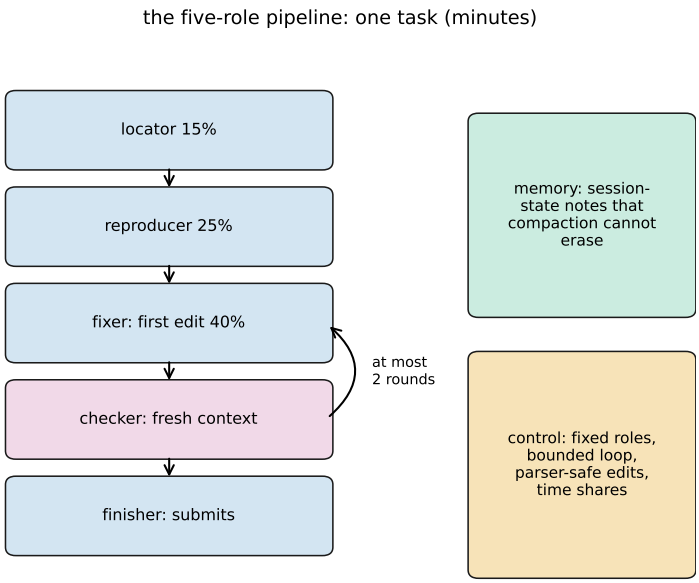

Figure 1: The five-role pipeline (entered on the leaderboard on 2026-10-07): five roles in a fixed order within one task; percentages are deadlines as shares of the time budget.